

Healthcare/IoT Cybersecurity Testbed: Sem. 2 Milestone 4

By: Justin Bower, Nathan Maloney, and Ipule Pipi

Instructors/Advisors: Dr. Sudhakaran and Dr. Aydeger

Testbed user

Local testbed — client machine

Device connection

USB / wireless capture
Adafruit BLE sniffer
EEPROM & chip readers

RE & analysis tools

Ghidra
Wireshark / Tshark
Binwalk & Foremost

Exploitation scripts

GDB / pwndbg (ret2*)
Pre-made & custom scripts

AWS cloud — server side

Web dashboard

React + Tailwind UI
Cognito authentication

API + compute

API Gateway
Lambda: instant + durable

Storage & sandbox

DynamoDB + S3 storage
Docker sandboxed analysis
CVE database

Uploads firmware, results & CVE data

System Architecture Explained

Main groups:

1. Prospective Users: (Security Researchers, Manufacturers, Engineers, and Advisors)
2. User-facing systems: Web Dashboard (Server-side) and Local GUI (Client-machine)
3. Internal Systems:
 - AWS Cloud backend (includes DynamoDB, S3 Storage, Backend API, and VM Compute)
 - Router with OpenWRT and Wireguard or Tailscale

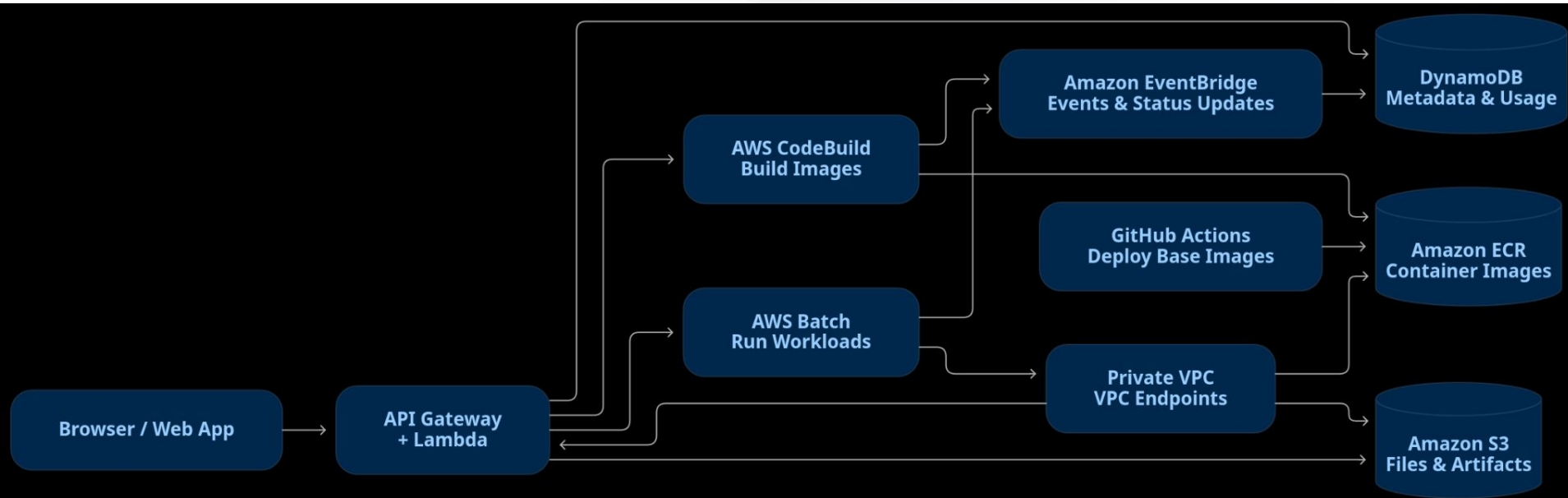
Milestone 4 Progress Summary

Task	% Complete	Justin	Nathan	Ipule	Jordan	To-Do
Project Management, Document Creation, and Overall Planning	100%	100%	0%	0%	0%	Continue managing project and creating documents as planning is now complete
Complete Firmware and Adafruit Demos	100%	33%	33%	33%	0%	Web interface demo of database functions
VR/RE & CVE database work	75%	50%	0%	0%	25%	Finish VR/RE of smart watches/heart monitors and enter any CVES
Local GUI development	50%	25%	0%	0%	25%	Add launch options, create more launch scripts, make GUIs launch in-frame

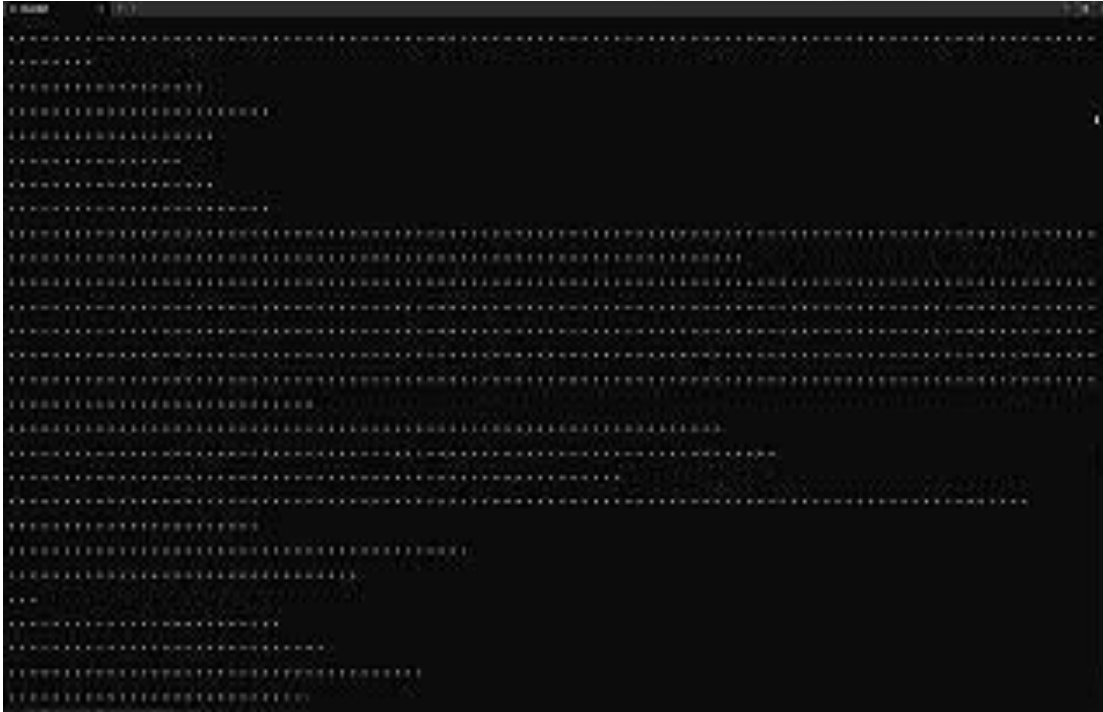
Milestone 4 Progress Summary

Testbed and Networking Set-up	100%	30%	0%	30%	30%	Flash the router with OpenWRT
Web Dashboard & Cloud Backend Integration for	0%	0%	60%	40%	0%	Integrate analysis initiation, job status, results/artifacts, workflow states, and authorization into dashboard
Cloud infrastructure & backend services	90%	0%	90%	0%	0%	Cloud API endpoints and firewall for local application
Artifact & Firmware Management	100%	0%	50%	50%	0%	Continue integration with analysis workflow

New Scripting Architecture



Adafruit Bluetooth Sniffer Demo



```

Directory: C:\Users\Justin\AppData\Roaming\Nordic Semiconductor\Sniffer\logs

Mode                LastWriteTime         Length Name
----                -
-a---              4/20/2026   7:54 PM         480130 lift_mouse_capture.pcap
-a---              4/20/2026   7:54 PM         572909 lift_mouse_log.txt

```

This preliminary demo shows step 1 of the whole workflow for Bluetooth RE

lift_mouse_capture.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

Interface phy0.mon Band 2.4 GHz Channel 1 · 2.412 GHz 20 MHz

No.	Time	Source	Destination	Protocol	Length	Info
19	0.902644			Packet	49	
20	0.943649			Packet	49	
21	0.976620			Packet	49	
22	0.978067			Packet	62	
23	0.979766			Packet	62	
24	0.981221			Packet	58	
25	0.982445			Packet	58	
26	0.983448			Packet	58	
27	0.984219			Packet	38	
28	0.985240			Packet	63	
29	0.986395			Packet	58	
30	0.987427			Packet	63	
31	1.309093			Packet	63	
32	1.411937			Packet	63	
33	1.413488			Packet	58	
34	1.414989			Packet	58	
35	1.417220			Packet	58	

Frame 35: Packet, 58 bytes on wire (464 bits), 58 bytes captured (464 bits)

- Encapsulation type: USER 10 (55)
- Arrival Time: Apr 20, 2026 19:52:51.134095000 EDT
- UTC Arrival Time: Apr 20, 2026 23:52:51.134095000 UTC
- Epoch Arrival Time: 1776729171.134095000
- [Time shift for this packet: 0.000000000 seconds]
- [Time delta from previous captured frame: 2.231000 milliseconds]
- [Time delta from previous displayed frame: 2.231000 milliseconds]
- [Time since reference or first frame: 1.417220000 seconds]
- Frame Number: 35
- Frame Length: 58 bytes (464 bits)
- Capture Length: 58 bytes (464 bits)
- [Frame is marked: False]
- [Frame is ignored: False]
- [Protocols in frame: user_dlt:data]
- Character encoding: ASCII (0)

User encapsulation not handled: DLT=157, check your Preferences->Protocols->DLT_USER

[Expert Info (Warning/Undecoded): User encapsulation not handled: DLT=157, check your Preferences->Protocols->DLT_USER]

Data (58 bytes)

```

0000 03 06 33 01 98 1b 06 0a 01 27 33 00 00 72 02 00  ..3.....r...
0010 00 d6 be 89 8e 60 20 ac 8c 64 6b cb da 02 01 05  ....dk....
0020 03 19 c2 03 05 03 12 18 72 fd 05 08 4c 49 46 54  ....r...LIFT
0030 06 ff 06 00 03 00 00 12 9d 60                                ....

```

lift_mouse_capture.pcap Packets: 7145

This screenshot shows step 2 of a Bluetooth RE workflow

This PCAP shows what the Adafruit sensor picks up when listening.

The frame captured information for a logitech wireless mouse

Physical Testbed



- This is still the current deployment of the physical testbed, featuring the virtual components, as follows:
 - Server-side -> Web Dashboard (Left Laptop)
 - Local-Side -> Analysis Tooling (Right Laptop)
- Router needs to be flashed with OpenWRT and a VPN tunnel with [Insert Chosen Application will be required.
- We will be integrating more devices as the RE/VR advances
- So far we have reverse engineered the oximeters and are now moving onto smart watches/heart monitors

Local GUI Progress



Milestone 5 Task Matrix

Task	Justin	Nathan	Ipule	Jordan
DS18B20 I2C Sensor/Buspirate Research and Integration	Spearhead Bus Pirate and sensor research/integration	Assist with the research and integration	Assist with the research and integration	Assist with the research and integration
Containerized Deployments and Platform Security Testing	Work with Jordan to port network tools to the GUI as needed and write deployment scripts for their cloud VM setup.	Assist with the creation and deployment of various docker containers	Assist with the creation and deployment of various docker containers	Work with Justin to port network tools to the GUI as needed and write deployment scripts for their cloud VM setup.
Start-to-Finish Demo	Verify integrations and create the demo	Create pipelines to allow for files to be easily used in each step of RE work	Create pipelines to allow for files to be easily used in each step of RE work	Create a firmware analysis / exploitation pipeline for extracted firmware
Project Management, Communication, and Documents	Manage project, documents, and communications,	Assist with document creation	Assist with document creation	Assist with document creation



Network Traffic Analysis Integration (Wireshark/Tshark)	Create & integrate Wireshark / Tshark scripts with Local GUI	Assist with testing	Assist with testing	Create & integrate Wireshark / Tshark scripts with Local GUI
Firmware RE & Analysis (Ghidra workflows, entropy analysis, Binwalk/Foremost segmentation)	Create & integrate Ghidra analysis / file-carving scripts with Local GUI	Assist with testing	Assist with testing	Create & integrate Ghidra analysis / file-carving scripts with Local GUI
WiFi/Bluetooth Exploitation Script Development	Continue development of pre-made exploitation scripts for wifi/bluetooth exploitation	Assist with research, implementation, and testing	Assist with research, implementation, and testing	Create & integrate WiFi/Bluetooth analysis scripts with Local GUI
Local GUI Development	Assist Jordan with research, implementation, and testing	Create API to allow local GUI to communicate with cloud securely	Assist with API	Design UI/UX & integrate tools into application
Real time file analysis on cloud platform	Test security with Nathan by ensuring proper file parsing with no code injection	Create systems for users to be able to run real time file analysis on the cloud	Create GUI that allows user to see the notebook and its results in real time	Test security with Nathan by ensuring proper file parsing with no code injection

**Thanks for
listening...
Questions?**